

EGS3012S (2024) Test 3B

Total: 25 (MCQ, ~70s/question) and 1 short Question

MCQ: Correct answer = +1, Incorrect = -0.25, Blank=0 (Negative marking)

Time Allowed: 45mins

Name:

Student No.:

#	Question	Answer
1	<u>In seasonal forecasting, the goal is to forecast daily weather throughout the season and not to forecast the seasonal averages.</u> TRUE or FALSE?	
2	<u>Climate feedbacks can either amplify warming or reduce it.</u> TRUE or FALSE?	
3	<u>Tropical Cyclones:</u> (A) Are characterised by clockwise rotating winds in the Northern Hemisphere and anti-clockwise rotating winds in the Southern Hemisphere (B) Do not occur at the Equator because the sea surface temperatures are too hot for tropical cyclone formation (C) Do not occur at the Equator due to there being zero coriolis force (D) Are strengthened when they move over cooler sea surface temperatures	
4	<u>What is the wavelength in degrees longitude of a Rossby Wave with zonal wavenumber $k=3$?</u> (A) 3° (B) 60° (C) 120° (D) 360°	
5	<u>Which of the following can trigger a thunderstorm?</u> (A) Waterspouts (B) Sea breeze fronts (C) Berg winds (D) Lightning	
6	<u>Which of the following is an example of climate?</u> (A) An intense thunderstorm in Johannesburg (B) A hot day in Cape Town in summer (C) The average temperature in Cape Town over the past 50 years (D) A foggy day in Cape Town	

7	<u>Climate models are used for which of the following purposes?</u> (A) Project future climate changes (B) Determine which factors are responsible for 20th century warming (C) Simulate past climate changes (D) All of the above	
8	<u>Which of the following is not an IPCC Representative Concentration Pathway (RCP) scenario?</u> (A) RCP3.5 (B) RCP4.5 (C) RCP 2.6 (D) RCP8.5	
9	<u>A necessary ingredient for a tornado to form is:</u> (A) Strong cyclonic rotation (B) Divergence of air at the surface (C) Sudden changes in land surface type (D) Strong vertical wind shear	
10	<u>Which of the following is NOT a teleconnection mode?</u> (A) Pacific Decadal Oscillation (PDO) (B) El Niño Southern Oscillation (ENSO) (C) West African Monsoon (WAM) (D) Indian Ocean Dipole (IOD)	
11	<u>Which of the following is correct about the weather forecasting process?</u> (A) Reanalysis data are combined with observations to initialise a new forecast (B) The output of a previous forecast is combined with reanalysis data to give analysis data (C) The output of a previous forecast is combined with observations to initialise a new forecast (D) Output of a previous forecast is combined with analysis data, the output of which is the new forecast	

12	<u>Surface weather maps show all of the following except:</u> (A) Precipitation amounts (B) Warm fronts (C) Sea level pressure distribution (D) Areas of high and low pressure	
13	<u>Air pollutant emissions can be from which of the following sources?</u> (A) Stationary (B) Anthropogenic (C) Gas (D) Natural (E) All of the above	
14	<u>Which of the points below is NOT a valid reason for parameterising a process in an atmospheric model?</u> (A) It is a sub-grid scale process (B) The process is too complex to represent (C) To simplify the model (D) We do not understand the process yet (E) None of the above; i.e. A--D are all valid reasons for parameterisation	
15	<u>Which of the statements below is NOT correct about atmospheric modelling?</u> (A) the horizontal resolution (grid size) should be at least 4 times bigger than the width of targeted atmospheric features (B) the time-step should be at least 4 times smaller than the lifespan of targeted atmospheric features (C) all sub-grid processes should be parameterized (D) none of the above	
16	<u>A probabilistic forecast is based on:</u> (A) A single model run (B) An ensemble of forecasts (C) A single bias-corrected model run (D) The mean of an ensemble of forecasts	

17	<u>Choose the correct answer</u> (A) Cut-off lows are always associated with heavy rainfall (B) Coastal lows are deep cyclonic systems that cause berg winds and fog (C) The easterly wave trough over South Africa is a summer feature associated with surface convergence to the east and divergence to the west (D) Subtropical anticyclones are associated with an elevated inversion which facilitates rainfall in the interior of South Africa (E) Particular synoptic types always cause the same weather at a particular location.	
18	<u>The largest anthropogenic driver of air pollution is</u> (A) Industrial activity (B) Domestic burning (C) Agricultural activities (D) Energy generation (E) Vehicles	
19	<u>Vehicle emissions are major primary sources of which of the following pollutants?</u> (A) O₃ (B) CO (C) PM₁₀ (D) None of the above	
20	<u>Choose the most correct statement:</u> (A) During the night time thermal turbulence dominates and pollutants are rapidly dispersed (B) During the day time thermal turbulence dominates and pollutants are mainly spread out horizontally closer to the ground (C) During the night time thermal turbulence dominates and pollutants are mainly spread out horizontally closer to the ground (D) During the day time thermal turbulence dominates and pollutants are rapidly dispersed	

21	<u>Which of the following pollutants impact negatively on both human health and ecosystems: (i) CO (ii) O₃ (iii) SO₂ (iv) NO₂</u> (A) i (B) ii, iii, iv (C) i, ii, iii (D) i, iii	
22	<u>In relation to a long-wave Rossby wave feature in the mid-latitude upper troposphere, a surface mid-latitude cyclone is most likely to occur?</u> (A) Along the ridge axis (B) West of the trough axis (C) Along the trough axis (D) East of the trough axis	
23	<u>Where, in the vicinity of a mid-latitude cyclone (MLC) are you likely to find the highest dew point temperature?</u> (A) At the MLC core (B) Aloft at the 500hPa level, directly above the cold front (C) Ahead of the surface warm front (D) In the near-surface warm sector immediately ahead of the cold front (E) In the near-surface cold sector, immediately behind the cold front	
24	<u>Surface anticyclones associated with an upper tropospheric jet streak tend to occur beneath which two quadrants of the jet streak?</u> (A) Equatorward entrance and exit quadrants (B) Poleward entrance and exit quadrants (C) Equatorward entrance and poleward exit quadrants (D) Poleward entrance and Equatorward exit zones	
25	<u>Which of the one following weather system types is a cold-cored closed low in the mid-troposphere?</u> (A) Cut-Off Low (B) Tropical Cyclone (C) Rossby Wave Ridge (D) Warm Front	

Section B (15 Marks)

Name:

Student No.:

7. In Figure 1 is a synoptic chart for a particular day of a particular month in a particular year at a particular time - this day is **Day 0**. The column Day +1 is the day after Day 0, meaning 24 hours later. Using your knowledge of South African weather systems draw in four to five contours of the **500hPa height** in the upper row for Day 0 and Day +1 and label regions of high and low pressure. In the bottom row draw in **sea level pressure** isobars for Day +1 and label regions of high and low pressure. On the **day 0 slp map** label the synoptic systems present. Lastly, fill in on the synoptic chart the likely month Day 0 occurred in. **[9 marks]**

8. On the table below are a set of variables and three cities for Day 0 and Day +1. Fill in values for each variable that would likely be recorded at the Airport of each city on each day. You are also required to suggest the dominant synoptic circulation type controlling the weather over the city. **[6 marks]**

Day 0

	Cape Town	Johannesburg
Tmax		
Tmin		
Wind speed		
Wind direction		
Rain (Yes/No)		
Synoptic system affecting the city		

Day +1

	Cape Town	Johannesburg
Tmax		
Tmin		
Wind speed		
Wind direction		
Rain (Yes/No)		
Synoptic system affecting the city		

